№ 1

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1. Octave
2.
3.
4.
5. 6

Octave

: sudo apt install octave
: 8.4.0

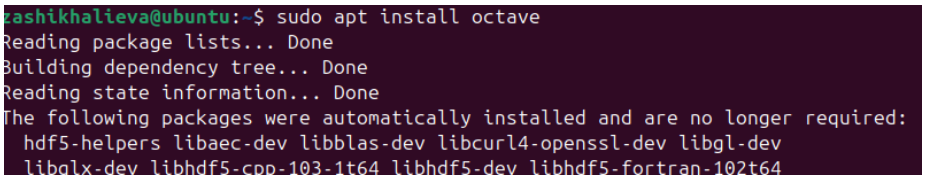

Figure 1:

:
y1 = sin x + (1/3) sin 3x + (1/5) sin 5x

y2 = cos x + (1/3) cos 3x + (1/5) cos 5x

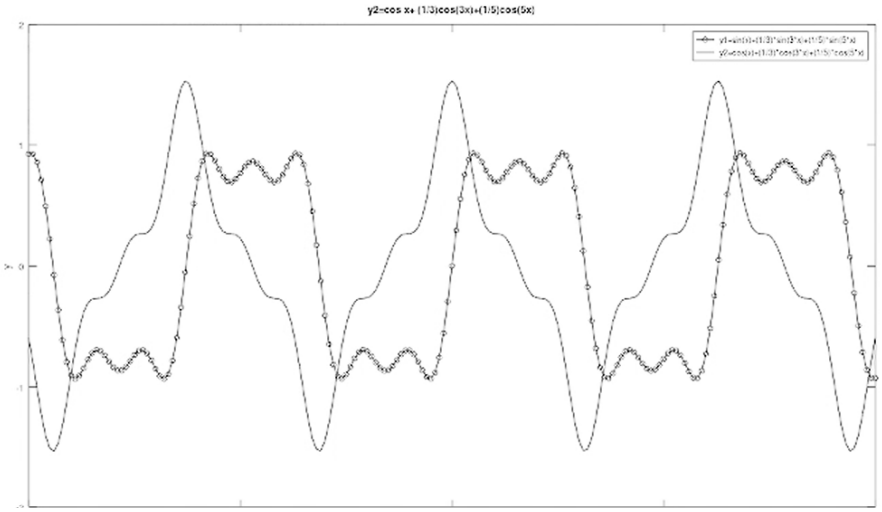

Figure 2:

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- ,

- 1: 10 , 1
- 2: 40 , 0.7

- : 5
- : 50

```
% количество отсчетов (гармоник):
N=8;
% частота дискретизации:
t=-1:0.01:1;
% значение амплитуды:
A=1;
% период:
T=1
% амплитуда гармоник
nh=(1:N)*2-1;
% массив коэффициентов для ряда, заданного через sin:
Am=2/pi ./ nh;
% массив гармоник:
harmonics=cos(2 * pi * nh' * t/T);
% массив элементов ряда:
s1=harmonics.*repmat(Am',1,length(t));
% Суммирование ряда:
s2=cumsum(s1);
% Построение графиков:
for k=1:N
subplot(4,2,k)
plot(t, s2(k,:))
end
% Экспорт рисунка в файл .png:
print("meander2.png");
```

Figure 3:

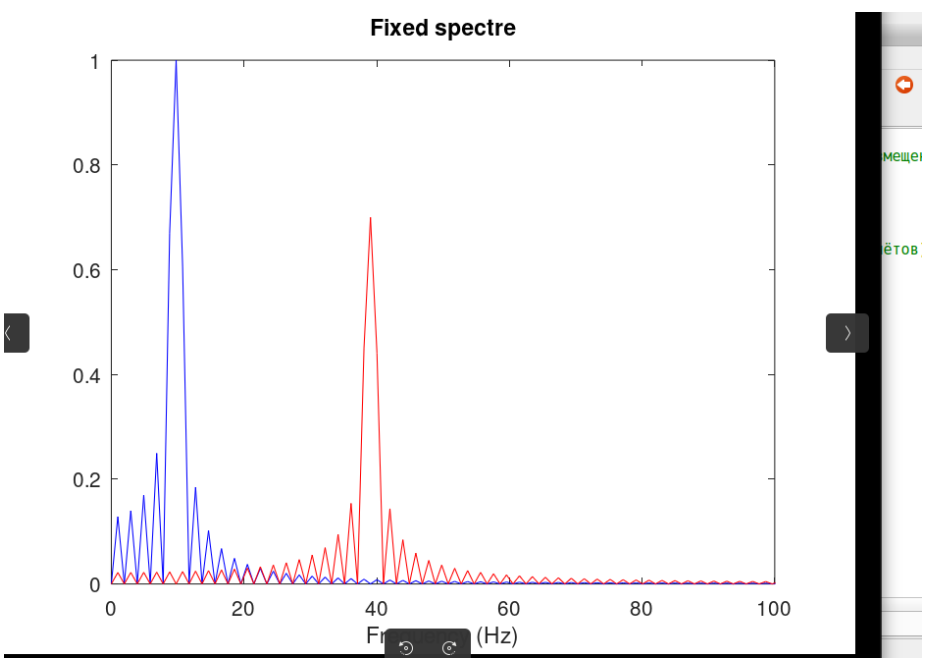

Figure 4:

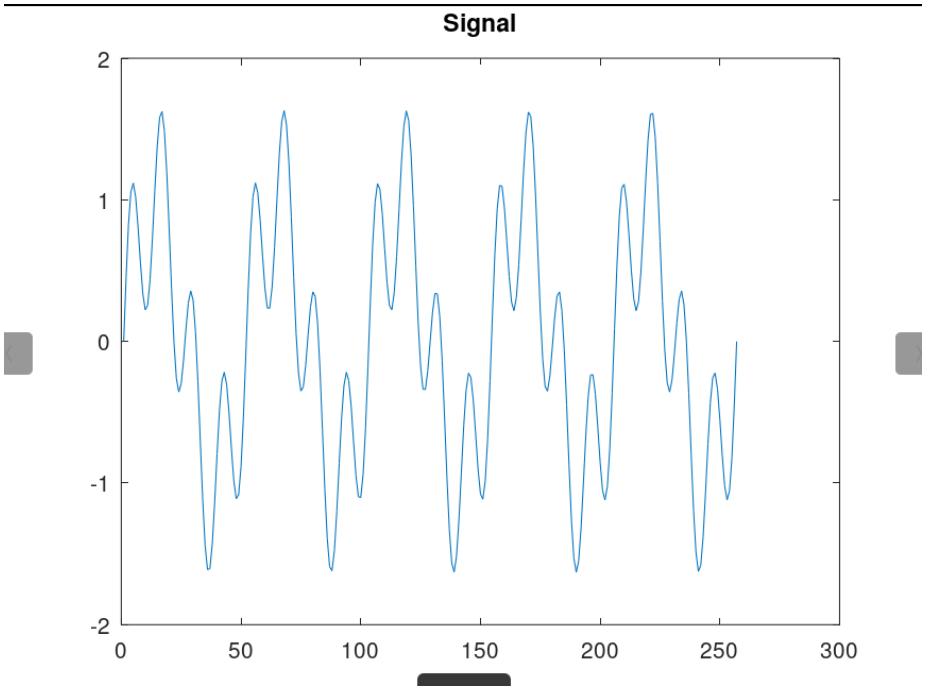

Figure 5:

	unipolar.m
AMI	ami.m
NRZ	bipolarnrz.m
RZ	bipolarrz.m
	manchester.m
	diffmanc.m

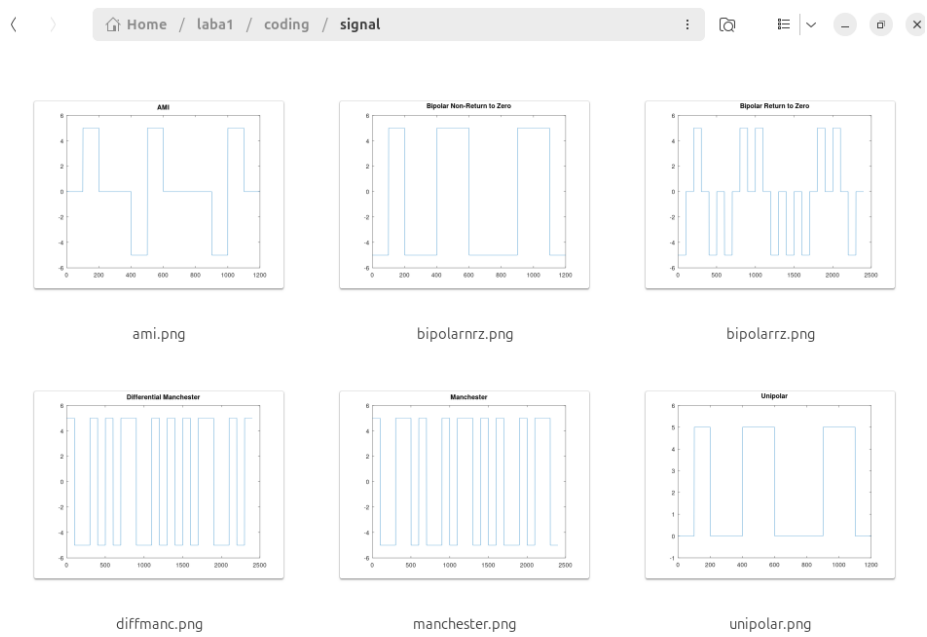

Figure 6:

AMI
NRZ

RZ

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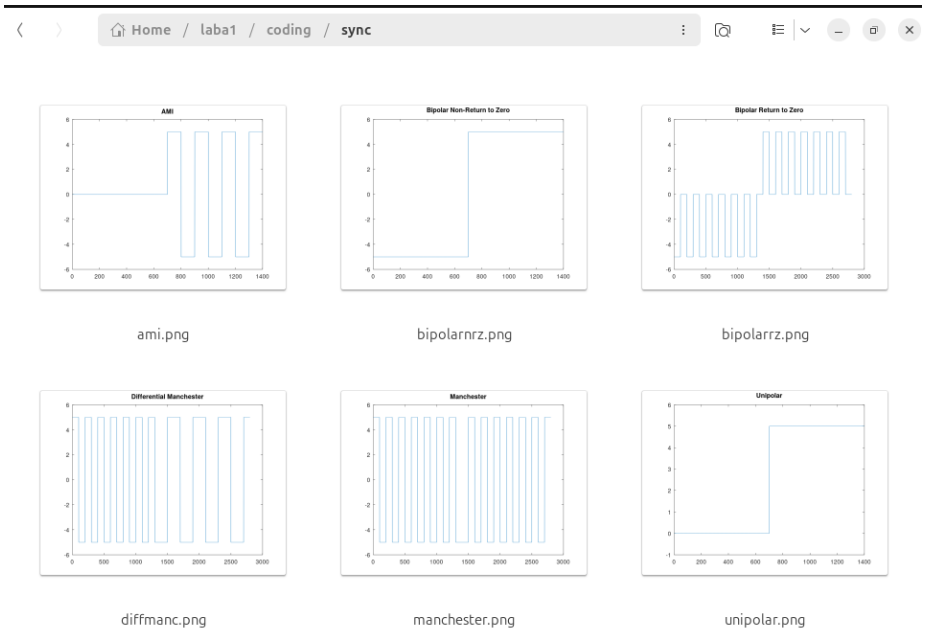

Figure 7:

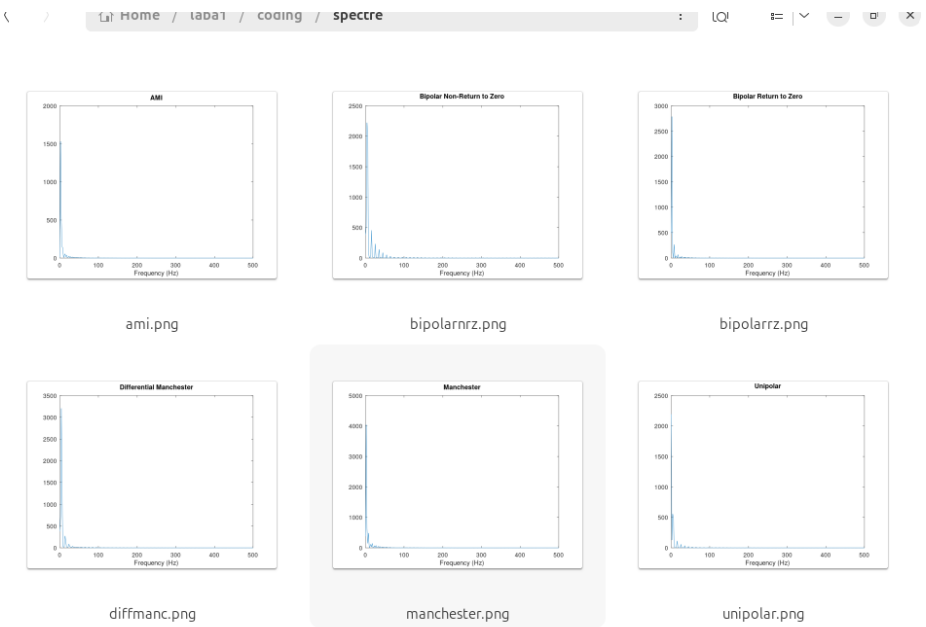

Figure 8: